Tutorial No : 7

Tutorial Title: Evaluation of postfix expression.
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Program:

```
#include <stdio.h>
#include <ctype.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int val) {
   stack[++top] = val;
}

int pop() {
   return stack[top--];
}

int evaluatePostfix(char* exp) {
   for (int i = 0; exp[i] != '\0'; i++) {

      if (isdigit(exp[i])) {
         push(exp[i] - '0');
      }

      else {
         int op2 = pop();
         int op1 = pop();

         switch (exp[i]) {
            case '+': push(op1 + op2); break;
            case '-': push(op1 - op2); break;
            case '*': push(op1 * op2); break;
            case '/': push(op1 / op2); break;
         }
      }
```

```
    }
    return pop();
}

int main() {
    char exp[] = "445*+3-";

    printf("Postfix Expression: %s\n", exp);
    printf("Result: %d\n", evaluatePostfix(exp));

    return 0;
}
```

Output:

